Sec: SR.IIT_*COSC(MODEL-A&APEX) | GTM-10 | Date: 19-03-26
Time: 3 Hrs | JEE-MAIN | Max. Marks: 300

KEY SHEET

PHYSICS

1	**B**	2	**C**	3	**A**	4	**D**	5	**C**
6	**D**	7	**B**	8	**C**	9	**B**	10	**A**
11	**B**	12	**D**	13	**B**	14	**C**	15	**C**
16	**D**	17	**B**	18	**B**	19	**D**	20	**C**
21	**6**	22	**200**	23	**400**	24	**4**	25	**202**

CHEMISTRY

26	**B**	27	**B**	28	**C**	29	**A**	30	**B**
31	**C**	32	**A**	33	**D**	34	**A**	35	**A**
36	**A**	37	**A**	38	**C**	39	**C**	40	**B**
41	**C**	42	**A**	43	**B**	44	**C**	45	**D**
46	**2**	47	**8**	48	**3**	49	**6**	50	**1**

MATHEMATICS

51	**C**	52	**A**	53	**B**	54	**C**	55	**B**
56	**C**	57	**A**	58	**D**	59	**B**	60	**B**
61	**B**	62	**B**	63	**B**	64	**C**	65	**B**
66	**C**	67	**C**	68	**C**	69	**B**	70	**A**
71	**18**	72	**5**	73	**4**	74	**12**	75	**3**

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SOLUTIONS

PHYSICS

1. The formula for g is $g = \frac{4\pi^2 L}{T^2}$

The percentage error is given by: % Error (g) = %Error (L) +2 × %Error (T)
%Error (g) = 0.4%+2 × (1.0%) = 0.4%
+2.0% = 2.4%

2. Slope of isothermal process: Slope of adiabatic process: Since (Reason R), the magnitude of the adiabatic slope is greater than the isothermal slope . Therefore, A is true and R is the correct explanation

3. (A) $\left(\upsilon_0 = \sqrt{\frac{GM}{R}} \propto R^{-1/2}\right)$ (Matches III)

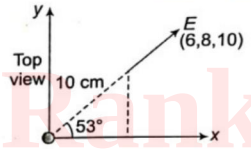
(B) Kepler's Law $\left(T^2 \propto R^3 \Rightarrow T \propto R^{3/2}\right)$ (Matches I)

(C) $L = mvr = m\sqrt{\frac{GM}{R}} \propto R^{\frac{1}{2}}$ (Matches IV)

(D) $\left(PE = -\frac{GMm}{R} \propto R^{-1}\right)$ (Matches II)

4. $E = \frac{\lambda}{2\pi\varepsilon_0 d} = \frac{2K\lambda}{d} = \frac{2\times 9\times 10^9 \times (10/9)\times 10^{-9}}{10\times 10^{-2}} = 200 V/m$

As field is radially out, there is no component of it in z direction.



$\vec{E} = \left(200\cos 50^0 \hat{i} + 200\sin 53^0 \hat{j}\right) V/m$

$= \left(200\times\frac{3}{5}\hat{i} + 200\times\frac{4}{5}\hat{j}\right) = \left(120\hat{i} + 160\hat{j}\right) V/m$

5. $X_L = \omega L = 2\pi fL = 220\Omega$

$\phi = \tan^{-1}\left(\frac{X_L}{R}\right) = \tan^{-1}\frac{220}{220} = 45°$

$I = \frac{E}{\sqrt{R^2 + X_L^2}} = \frac{220}{\sqrt{(220)^2 + (220)^2}} = \frac{1}{\sqrt{2}}$

Wattless component of current = $I \sin\phi$

$= \frac{1}{\sqrt{2}}.\sin 45^0 = 0.5A$

6. $a = \frac{v\,dv}{dx}, \frac{dv}{dx} \rightarrow$ slope; So, $\frac{-4}{2} = -2$

Intercept = + 4

a $\rightarrow$ negative; So, $a- = \frac{v\,dv}{dt} = 4x - 8$

7. $R = R_0 (A)^{1/3}$

$S = 4\pi R^2 \Rightarrow S \propto (A)^{2/3}$

$A_1 = 8, A_2 = 27$

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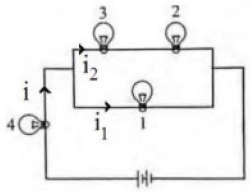
$$\frac{S_2}{S_1}=\left(\frac{27}{8}\right)^{2/3}=\frac{9}{4}=2.25$$

8. $\frac{3}{4}I_0=I_0\cos^2\left(\frac{\phi}{2}\right)\Rightarrow\frac{\phi}{2}=\frac{\pi}{6}\Rightarrow\phi=\frac{\pi}{3}$

$$\frac{\pi}{3}=(d\sin\theta)\Rightarrow\sin\theta=\frac{\lambda}{6d}$$

9. $K=\frac{p^2}{2m},p=\frac{h}{\lambda}$

10.



$$i_2^2R_2=i_2^2R_3\Rightarrow R_2R_3$$

$$\therefore\frac{v_1^2}{4R_2}=\frac{v_1^2}{36}\Rightarrow R_2=9\Omega$$

$$\frac{i_1}{i_4}=\frac{1}{2}\Rightarrow i^2R_4=\frac{i^2}{9}36$$

$$\therefore R_4=9\Omega$$

11. $Av=Av'+AV'\Rightarrow v'=\frac{v}{2}$

$$F=\frac{\Delta\rho}{\Delta t}=av^2\rho+A\frac{v^2}{4}\rho$$

12. $Y=A\left(\overline{AB}\right)=A.\left(\overline{A}+\overline{B}\right)=A\overline{A}+A\overline{B}$

$$\therefore Y=A\overline{B}$$

13. Distance of first image (I_1) formed after refraction from the plane surface of water is $\frac{10}{4/3}=7.5cm$, from water surface $\left(d_{app}=\frac{d_{actual}}{\mu}\right)$

Now distance of image $=5+7.5=12.5cm$ from the plane mirror.

14. $d=x\left[\frac{p-q}{p+q}\right]$

15. Using perpendicular axis theorem

$I_Z=I_x+I_y$ by symmetry $I_x=I_y$ and I_z is the same

16. $W=\mu mgR$

17. $V=\frac{KQ}{R_1}$

$$V^1=\frac{KQ}{R_2}$$

18. $\Delta U=bpdv=b(nRdT)$

$$\Delta w=bdv=nRdT$$

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$\Delta Q = \Delta U + \Delta W$

$nCpdT = nRdT(b) + nRdT$

$C_p = (b+1)R \Rightarrow C_v = bR$

$\gamma = \frac{C_p}{C_v}$

19. $n = \frac{(2p+1)}{41}; 1250 = \frac{(2p+1)340}{4\times 85}\times 100$

$2P+1 = 12.5$

It is in between 11th and 13th harmonic

13th harmonic

13th harmonic is not possible

No.of resonance =6

$[1,3,5,7,9,11]$

21. $\frac{n(n-1)}{2}$

23. $a = v\left(\frac{dv}{dx}\right) = 4-2x$

24. $\frac{I_0}{2}\cos^2\theta$

CHEMISTRY

26. Given the given graph 2 ml NaOH solution is used for neutralization of HCl and 3 ml NaOH solution is used for neutralization of CH_3COOH

Mole of HCl = Mole of NaOH used

M × 40= 0.1× 2

M = 0.005

Mole of CH_3COOH = Mole of NaOH used

M ×40 = 0.1× 3

M = 0.0075

HCl is strong acid and will be neutralized first.

27. m.eqts of excess $H_2SO_4 = 10\times 0.1 = 1$

m.eqts of H_2SO_4 taken $= 50\times 0.05\times 2 = 5$

m.eqts of H_2SO_4 reacted with $NH_3 = 4 =$ m.eqts of $(NH_4)_2C_2O_4$

wt of $(NH_4)_2C_2O_4 = 4\times 62\times 10^{-3}$

% purity $= \frac{4\times 0.062}{}\times 100 = 50\%$

28. $= \lambda_1 = \sqrt{\frac{150}{100}}; \lambda_2 = \sqrt{\frac{150}{81}}; \lambda_3 = \sqrt{\frac{150}{49}}$

29. Increasing order of Dipole moment

$NF_3 < HBr < H_2S < CHCl_3$

$\mu = 0.24D \;\; 0.79D \;\; 0.95D \;\; 1.04D$

It is NCERT based

30. Compare the structures of CO, CO_2, CO_3^{-2} strength of the back bonding decreases in halogens from top to bottom

32. $\Delta_f S^0(NH_4Cl, s)$ at 300K

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$$= S^0_{NH_4Cl(s)} - \left[\frac{1}{2}S^0_{N_2} + 2S^0_{H_2} + \frac{1}{2}S^0_{Cl_2}\right]$$

$$= -374 JK^{-1}mol^{-1} \because \Delta_r C_p = 0$$

$$\because \Delta_f S^0_{310} = \Delta_f S^0_{300} = -374 JK^{-1}mol^{-1}$$

$$\Delta_f H^0_{310} = \Delta_f S^0_{300} = -314.5$$

$$\Delta_f G^0_{310} = \Delta_f H^0 - 310\Delta S^0$$

$$= -314.5 - \frac{310(-374)}{1000} = -198.56 kj/mol$$

33. $N_2O_5(g) \rightleftharpoons N_2O_3(g) + O_2(g), K_c = 2.5$

At equ. $4-x$ $x-y$ $x+y$

$N_2O_3(g) \rightleftharpoons N_2O(g) + O_2(g)$

At equ $x-y$ y $(x+y)$

Give that $x + y = 2.5$ ……..(1)

$$K_c = \frac{[N_2O_3][O_2]}{[N_2O_5]} = \frac{(x-y)2.5}{4-x} = 2.5$$

$\Rightarrow x - y = 4 - x$

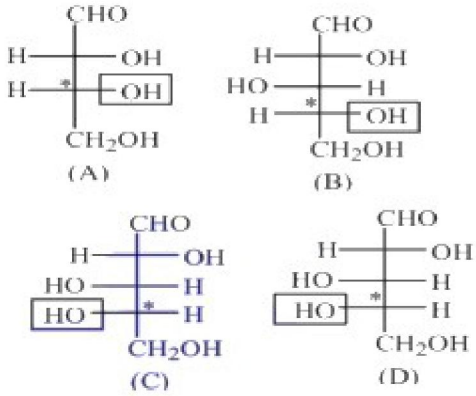
$\Rightarrow 2x - y = 4$ ……(2)

Solving (1) and (2) $\frac{x+y=2.5}{3x=6.5} \Rightarrow x = \frac{6.5}{3}$

$\therefore$ At equilibrium $[N_2O_5] = 4 - x = 4 - \frac{6.5}{3} = \frac{5.5}{3} = 1.83M$

34. $\frac{x}{y} = \frac{2}{1}$

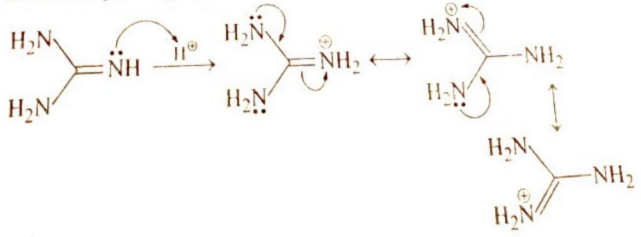
35.



In A, B, D* – OH group in right hand side then D-configuration is assign

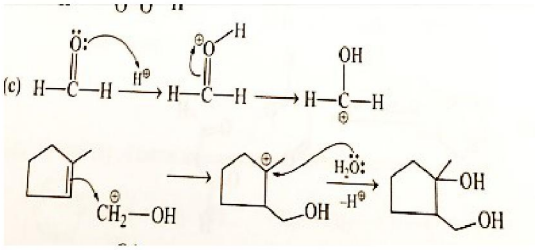
36 .

(a) Cation of (*p*) is highly resonance stable so it is most basic



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38.



39. Ionic radii $-Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$

Ionisation energy $-B < C < O < N$

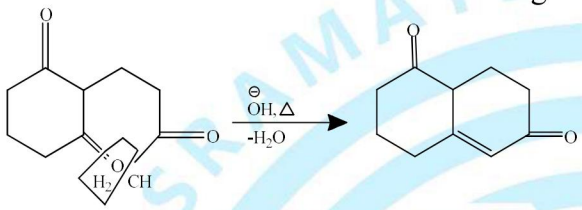
Metallic character $-B < Al < Mg < K$

Electron negativity $-Si < P < S < Cl$

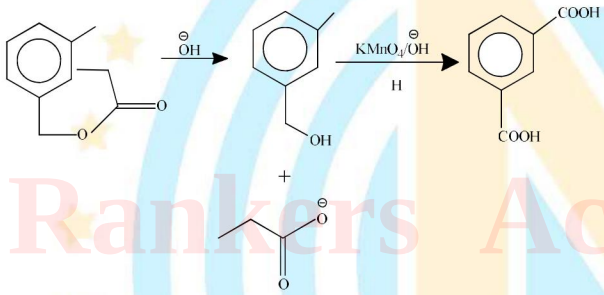
40. Based on number of alpha hydrogen atoms.

41. Reimer Tiemann's reaction.

42. Aldol condensation. 6 or 5 membered ring are obtained by intra-molecular reaction.

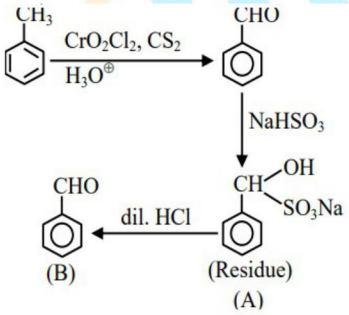


43.

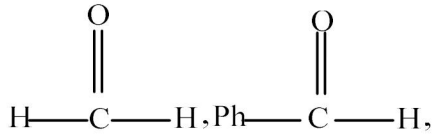


44. The elements present in the compound are converted from covalent form into ionic form by fusing the compound with sodium in Lassigne's test

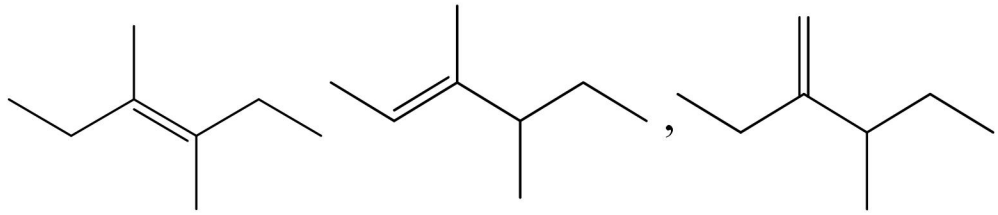
45.



46.

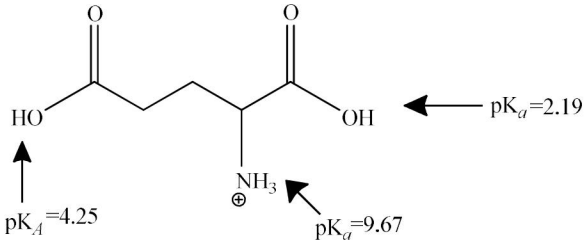


47.



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48.

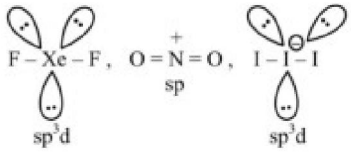


$$pI = \frac{4.25+2.19}{2} = 3.22$$

49. Linear species are

$Cl-Be-Cl$, $O=C=O$, $N^-=N^+=N^-$

(sp) (sp) (sp)



50. $\Delta T_f = iK_f m$

$0.0558 = i \times 1.86 \times 0.01$

$\Rightarrow \quad i = 3$

Structure should be:

$\left[Co(NH_3)_5 Cl\right]Cl_2$

MATHS

51. $g(x) = \int \frac{4x^7 + 6x^5}{\left(1 + x^2 + 3x^6\right)^2} dx$

Substitute

$t - x^{-6} + x^{-4} + 3 \Rightarrow dt = \left(-6x^{-7} - 4x^{-5}\right)$

$g(x) = \int \frac{-dt}{t^2} = \frac{1}{t} + C = \frac{x^6}{3x^6 + x^2 + 1} + C$

Since $\lim_{x \to 0} g(x) = 0, C = 0$ Thus,

$g(1) = \frac{1}{3+1+1} = \frac{1}{5}$

For matrix A, expanding along the top row:

$|A| = 1\left(5\left(\frac{1}{5}\right) - \alpha^2\right) - \alpha^2$

We know $|B| = |adj(adjA)| = |A|^4$

$|A|^4 = 16 \Rightarrow |A| = \pm 2$

Taking

$|A| = -2 \Rightarrow 1 - \alpha^2 = -2 \Rightarrow \alpha^2 = 3$

52. $f(t) = -(t-1)^2 + \frac{1}{4}$. The maximum value is $\frac{1}{4}$, so $e = \frac{1}{4} \Rightarrow e^2 = \frac{1}{16}$

Latus rectrum $= \frac{2b^2}{a} = 15 \Rightarrow b^2 = \frac{15a}{2}$

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Also, $b^2 = a^2\left(1-e^2\right) = \frac{15a^2}{16}$

Equating them: $\frac{15a}{2} = \frac{15a^2}{16} \Rightarrow a = 8$

Substitute $a = 8$ to get $b^2 = 60$

Thus $a^2 + b^2 = 64 = 124$

53. Let the parametric coordinates of Q on the line be $\left(1 + r\cos\alpha, 2 + r\sin\alpha\right)$ where $r = \sqrt{\frac{8}{3}}$

Substitute into $x + y = 5$

$$\left(1 + \sqrt{\frac{8}{3}}\cos\alpha\right) + \left(2 + \sqrt{\frac{8}{3}}\sin\alpha\right) = 5$$

This simplifies to $\cos\alpha + \sin 2\alpha = \frac{1}{2}$

The possible angles are $2\alpha_1 = \frac{\pi}{6}$ and

$2\alpha_2 = \frac{5\pi}{6}$, giving $\alpha_1 = \frac{\pi}{12}$ and $\alpha_2 = \frac{5\pi}{12}$

Their sum is $\frac{6\pi}{12} = \frac{\pi}{2}$

54. Let $\vec{\upsilon} = x\vec{a} + y\vec{b} = x\hat{i} + (x+y)\hat{j} + y\hat{k}$

Since $\vec{\upsilon}.\vec{a} = 0$ we have

$x(1) + (x+y)(1) = 0 \Rightarrow y = -2x$

So, $\vec{\upsilon} = x\left(\hat{i} - \hat{j} - 2\hat{k}\right)$

The projection on $\vec{c}$ is $\frac{|\vec{\upsilon}.\vec{c}|}{|\vec{c}|} = \sqrt{6}$

$|\vec{c}| = \sqrt{1^2 + 2^2 + (-1)^2} = \sqrt{6}$

$\frac{|x(1-2+2)|}{\sqrt{6}} = \sqrt{6} \Rightarrow |x| = 6$

Thus

$|\vec{\upsilon}|^2 = 36\left(1^2 + (-1)^2 + (-2)^2\right) = 36(6) = 216$

55. Le the terms be

$A-3d, A-d, A+d, A+3d$ (common difference 2d)

Sum $= 4A = 24 \Rightarrow A = 6$

Product

$+(2d)^4 = 961 \Rightarrow \left(36 - 9d^2\right)\left(36 - d^2\right) + 16$

$25d^4 - 360d^2 + 1296 = 961 \Rightarrow 5d^4 - 72d^2$

$\left(5d^2 - 67\right)\left(d^2 - 1\right) = 0$

Since terms are intergers, $d^2 = 1 \Rightarrow d = \pm 1$

(commom difference =2)

Terms are 3, 5, 7, 7. Largest term =9

56. Let $P\left(t^2, 2t\right)$ be a point on $y^2 = 4x$. Let its image in $x + y - 2 = 0$ be $Q(h,k)$

Using the image formula:

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$$\frac{h-t^2}{1}=\frac{k-2t}{1}=\frac{-2\left(t^2+2t-2\right)}{1^2+1^2}=-\left(t^2+\right)$$

$$h=-2t+2\Rightarrow t=\frac{2-h}{2}$$

$$k=-t^2+2\Rightarrow k=-\left(\frac{2-h}{2}\right)^2+2\Rightarrow\left(h-\right)$$

Comparing with $(x-a)^2=b(y-c)$ we get $a=2,b=-4.c=2$

57. Let $a=\frac{4}{7},b=\frac{1}{3}$

Multiply N^r and D^r by $(a-b)=\frac{4}{7}-\frac{1}{3}=\frac{5}{21}$

$$\frac{1}{a-b}\left[\left(a^2-b^2\right)+\left(a^3-b^3\right)+\left(a^4-b^4\right)+.....\infty\right]$$

$$\frac{1}{a-b}\left[\frac{a}{1-a}-\frac{b}{1-b}\right]=\frac{21}{5}\left[\frac{\frac{16}{49}}{1-\frac{4}{7}}-\frac{\frac{1}{9}}{1-\frac{1}{3}}\right]$$

$$=\frac{21}{5}\left[\frac{16}{21}-\frac{1}{6}\right]=\frac{21}{5}\left[\frac{96-21}{21.6}\right]$$

$$=\frac{75}{5.6}=\frac{15}{6}=\frac{5}{2}$$

58. Given $b_{ij}=2^{i+j-2}a_{ij}$

Factoring out powers of 2 from rows and columns of $|B|$:

$$32=64|A|\Rightarrow|A|=\frac{1}{2}$$

59. Words starting with A: $\frac{5!}{1!}=120$

Words starting with B H A A:2!=2

Words starting with B H A R A T; 1

Rank = 120 + 24 + 2 + 1 =147

60. $45^n=\left(3^2+5\right)^n=3^{2n}\times5^n$

Exponents of 3 in

$$80!=\left\lfloor\frac{80}{3}\right\rfloor+\left\lfloor\frac{80}{9}\right\rfloor+\left\lfloor\frac{80}{27}\right\rfloor=26+8+2=36$$

So $2n\le36\Rightarrow n\le18$

$$80!=\left\lfloor\frac{80}{5}\right\rfloor+\left\lfloor\frac{80}{25}\right\rfloor=16+3=19$$

The limiting factor is $n\le18$. Thus, maximum $n=18$

61. L_1 passes through $\vec{a}_1(1,1,1)$ with direction $\vec{b}_2(\alpha,1,\alpha)$.

$\vec{a}_2-\vec{a}_1=(0,-1,1)$ $\vec{b}_1\times\vec{b}_2=\hat{i}(-\alpha)-\hat{j}(2\alpha^2)+\hat{k}(3\alpha)$

Shortest distance

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$$d=\frac{\left|(\vec{a}_2-\vec{a}_1).(\vec{b}_1\times\vec{b}_2)\right|}{\left|\vec{b}_1\times\vec{b}_2\right|}=\frac{\left|2\alpha^2+3\alpha\right|}{|\alpha|\sqrt{4\alpha^2+10}}=$$

Setting $d=\frac{1}{\sqrt{2}}$

$2(2\alpha+3)^2=4\alpha^2+10\Rightarrow\alpha^2+6\alpha+2=0$

The sum of the roots $\alpha_1+\alpha_2=-6$

62. For $g(0)\Rightarrow\lambda=0\Rightarrow y=|-4|-|2|=2$ Area $g(0)=\int_0^1\left(2-\sqrt{x}\right)dx=2-\frac{2}{3}=\frac{4}{3}$

For $g(2)\Rightarrow\lambda=2\Rightarrow|2x-4|-|2-2x|$

Since $x\in(0,1)$, $2x-4<0$ and $2-2x>0$

$y=-(2x-4)-(2-2x)+2x^2=2+2x^2$

Area

$$g(2)=\int_0^1\left(2+2x^2-\sqrt{x}\right)dx=\left[2x+\frac{2x^3}{3}\right]\ g(0)+g(2)=\frac{4}{3}+2=\frac{10}{3}$$

63. $-1\le\frac{2}{x^2-x-1}\le1$

Condition 1

$\frac{2}{x^2-x-1}\le1\Rightarrow\frac{x^2-x-3}{x^2-x-1}\ge0$

Condition 2

$\frac{2}{x^2-x-1}\ge-1\Rightarrow\frac{x^2-x+1}{x^2-x-1}\ge0$

Solving the intersection of these inequalities gives the domain

$$\left(-\infty,\frac{1-\sqrt{13}}{2}\right]\cup\left[\frac{1+\sqrt{13}}{2},\infty\right)$$

64. For $X=\{1,2.....13\}$ Mean $\bar{x}=7$

Variance $\sigma_x^2=\frac{13^2-1}{12}=14$

For $Y=ax+b$ Mean $=7a+b=20$

Variance

$=a^2\sigma_x^2=14a^2=126\Rightarrow a^2=9\Rightarrow a=\pm3$

If $a=3,21+b=20\Rightarrow b=-1$

If $a=-3,21+b=20\Rightarrow b=41$

Sum of values of $b=-1+41=40$

65. $f(x)=\left|e^x-1\right|-|x|$

For $x>0$

$f(x)=1-e^x+x\Rightarrow f(x)=-e^x+1>0$

Increasing statement (III) if FLASE

At $x=0$, $RHD=e^0-1=0$ and $LHD=-e^0+1=0$. Since $LHD=RHD$, f is differentiable at $x=0$. Statemetn (I) is TRUE

66. Apply L'Hospital's rule on the limit:

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$$\lim_{t\to x}\frac{2ty(x)-x^2y(t)}{-1}=2\Rightarrow x^2y(x)-2xy(x)$$

Integrating factor $(I.F)=e^{\int\frac{-2}{x}}=\frac{1}{x^2}$

$$\frac{d}{dx}\left(\frac{y}{x^2}\right)=\frac{2}{x^4}\Rightarrow\frac{y}{x^2}=-\frac{2}{3x^3}+C$$

$$y(1)=1\Rightarrow 1=-\frac{2}{3}+C\Rightarrow C=\frac{5}{3}$$

$$y(2)=-\frac{1}{3}+\frac{20}{3}=\frac{19}{3}\Rightarrow 3y(2)=19$$

67. We have $2f(x)+3f\left(\frac{10}{x}\right)=4x$

Replace x with $\frac{10}{x}$: $3f(x)+2f\left(\frac{10}{x}\right)=\frac{40}{x}$

Multiply the first by 2 and the second by 3, then subtract:

$$(9-4)f(x)=\frac{120}{x}-8x\Rightarrow 5f(x)=\frac{120}{x}-8$$

$$5f(2)=60-16=44$$

$$5f(5)=24-40=-16$$

$$5f(2)-5f(5)=44-(-16)=60$$

68. Let $t=x^2$. The equation $t^2-kt+16=0$ must have two distinct positive roots.
Discriminant $\Delta=k^2-64>0\Rightarrow k>8$
Product of root $=16>0$ and sum $=k>0$
The smallest integer $=k>8$ is $k=9$

69. $\left(\vec{a}+2\vec{b}\right)\times\vec{c}=\vec{0}\Rightarrow\vec{c}=\lambda\left(\vec{a}+2\vec{b}\right)$

$$\vec{a}+2\vec{b}=(5,-1,3)\Rightarrow\vec{c}=\lambda(5,-1,3)$$

$$\left(\vec{a}+2\vec{b}\right)\cdot\vec{c}=(3,0,1)\cdot\lambda(5,-1,3)=\lambda(15+3)$$

Given $18\lambda=18\Rightarrow\lambda=1$
Therefore, $\vec{c}=(5,-1,3)$

$$|\vec{c}|^2=25+1+9=35$$

70. $f(0)=a$

$$LHL=\lim_{x\to0^-}\frac{1-\cos(bx)}{x^2}=\frac{b^2}{2}$$

$$RHL=\lim_{x\to0^-}\frac{\tan x-\sin x}{x^3}=\frac{1}{2}$$

Continuity implies $LHL=RHL=f(0)$

$\frac{b^2}{2}=\frac{1}{2}\Rightarrow b^2=1$. And $a=\frac{1}{2}$

$$a^2+b^2=\frac{1}{4}+1=\frac{5}{4}$$

71. $f(x)=1+x\int_0^1(y)\,dy+\int_0^1y^2f(y)\,dy$

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Let $A=\int_0^1 (y)\,dy$ and $B=\int_0^1 y^2 f(y)\,dy$

Thus $f(x)=1+Ax+B$

$A=\int_0^1 (1+Ay+B)\,dy=1+\frac{A}{2}+B\Rightarrow A$

$B=\int_0^1 y^2(1+Ay+B)\,dy=\frac{1}{3}+\frac{A}{2}+\frac{B}{3}\Rightarrow A$

Solving these simultaneious equations gives

$B=5$ and $A=12$

$f(x)=1+12x+5=5=12x+6$

$f(1)=18$

72. Let $h=3\cos\theta$ and $k=3\sin\theta$

The point Q is $(9\cos\theta-1,6\sin\theta+1)$

Locus if found by rearranging: $\frac{x+1}{9}=\cos\theta$ and

$\frac{y-1}{6}=\sin\theta\Rightarrow\frac{(x+1)^2}{81}+\frac{(y-1)^2}{36}=1$

This is an ellipse with $a^2=81$, $b^2=36$

Eccentricity $e^2=1-\frac{36}{81}=\frac{5}{9}$

Thus , $9e^2=5$

73. For $n=1$, product $=2$

For $n=2$, product $=2\times5=10$

For $n=3$, product $=10\times10=100$

For $n=4$, product $=100\times17=1700$

Thus $n=4$

74. For each of the 4 elements is S, there are 4 ways to distribute it between subsets A and B:

(In A , In B), (In a, Not B), (Not A, In B), (Not A, Not B)

To satisfy $A\cup B=S$, the elements cannot be in the (Not A, Not B) state. This leaves exactly 3 Valid choices per element.

Total favorable pairs $=3^4$. Total possible pairs in $P(S)\times P(S)=4^4=2^8$

Probability $=\frac{3^4}{2^8}$

Comparing to $=\frac{3^p}{2^q}$, $p=4$ and $q=8$. Thus $p+q=12$

75. $x=\frac{\pi}{12},\frac{\pi}{3},\frac{7\pi}{12}$

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